

OMKAR GURU

La Jolla, California | (858) 228-7232 | oguru@ucsd.edu

EDUCATION

UC San Diego

- 2029

B.S. Computer Science, GPA: 4.0

Geetanjali Olympiad School

- 2025

Grade XII - 94%

EXPERIENCE

Google Gemini API Developer Competition

- 2024

- Built a voice-based AI chatbot designed to make 2,980 government schemes easily accessible to everyone in any of India's vernacular languages.
- Used selenium to web scrape the schemes and used an RAG architecture to pass the context to the LLM
- Used PyAudio and Google's speech to text API to enable voice based input in vernacular languages.

International Journal of Science and Research Archive

- 2024

- Compared the performance of different architectures and embeddings for conversational assistants using large language models. Link: <https://ijsra.net/content/need-retrieval-augmented-generation-conversational-assistants-government-social-welfare>
- Concluded that RAG substantially improves the quality of the models predictions.

Intel, Inc. - AI for Youth Program

- 2022

- Completed, hands-on labs and projects. Trained an AI model to predict fake news and classify its sentiment using sentiment analysis. Compared the performance between a Decision Tree and KNN architecture for this use case.

AWS Young Builder's Challenge

- 2021

- Trained and deployed a regression model for predicting cardiac arrests based on medical data.

Personal Project - Frame Interpolation and Monocular Depth Estimation

- 2022

- Set up RIFE for frame interpolation and MIDAS for monocular depth estimation to run using cuDNN on GCP

SKILLS

Python: Familiar with packages like NumPy, PANDAS, OpenCV, PyTorch, Langchain, BeautifulSoup, FastAPI, ChromaDB, PyAudio, pynput, Scikit-learn, Matplotlib, SciPy, PyMySQL

Other Languages: SQL, Java, C++, C

Other Skills: Built multiple computers and serviced electronics like desktops, laptops, Nintendo Switch controllers. Comfortable debugging hardware and software. Amateur video editor and colorist.